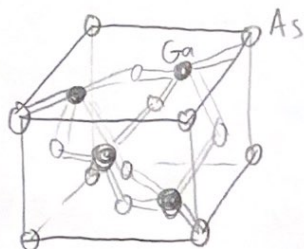


EE-232 Home work #1

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1.



GaAs has a zinc-blend structure.

There are 4 Ga atoms

$$8 \cdot \frac{1}{8} + 6 \cdot \frac{1}{2} = 4 \text{ As atoms in one cell.}$$

$$\left. \begin{array}{l} M_{\text{Ga}} = 1.157 \times 10^{-22} \text{ g} \\ M_{\text{As}} = 1.244 \times 10^{-22} \text{ g} \end{array} \right\} \text{atomic weight} \quad \text{and} \quad \left. \begin{array}{l} a = 5.653 \text{ \AA} \\ = 5.653 \times 10^{-8} \text{ cm} \end{array} \right\} \text{lattice constant}$$

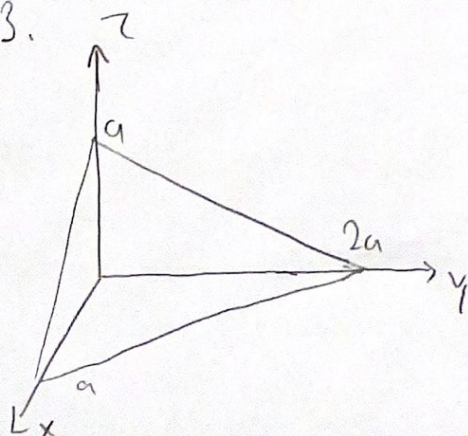
$$\rho = \frac{4 M_{\text{Ga}} + 4 M_{\text{As}}}{a^3} = \frac{4 \times 10^{-22} (3.401) \text{ g}}{(5.653 \times 10^{-8} \text{ cm})^3} = \boxed{4.53 \text{ g/cm}^3}$$

2. $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{\langle 110 \rangle \cdot \langle 111 \rangle}{|\langle 110 \rangle| |\langle 111 \rangle|} = \frac{2}{\sqrt{2} \cdot \sqrt{3}} = \frac{2}{\sqrt{6}}$

normal vectors

$$\theta = \arccos(2/\sqrt{6}) = \boxed{35.264^\circ}$$

3.



axis intercepts $\rightarrow a, 2a, a$

take reciprocal $\rightarrow 1, \frac{1}{2}, 1$

multiply with 2 $\rightarrow 2, 1, 2$

$$\boxed{(2, 1, 2) \text{ plane}}$$

$$4. \quad E_n = -\frac{E_0}{n^2}, \quad \Delta E = E_4 - E_1 = -13.6 \left(\frac{1}{4^2} - \frac{1}{1^2} \right) \text{ eV}$$

↓
For the hydrogen atom

$$\Delta E = +13.6 \times \frac{15}{16} \text{ eV} = 12.75 \text{ eV}$$

$$\Delta E (\text{in eV}) = \frac{1.24}{\lambda (\text{in } \mu\text{m})} \quad \lambda = \frac{1.24 \text{ eV}}{12.75 \text{ eV}} \mu\text{m} = \boxed{0.097 \mu\text{m} = 9.7 \times 10^{-8} \text{ m}}$$

$$5. \quad E = h\nu = \frac{hc}{\lambda} \quad E_{400\text{nm}} = 6.626 \times 10^{-34} \text{ J.s} \times \frac{3 \times 10^8 \text{ m/s}}{400 \times 10^{-9} \text{ m}}$$

$$= 4.97 \times 10^{-19} \text{ J} = 3.1 \text{ eV},$$

Energy range: $\boxed{(1.77 \text{ eV} - 3.1 \text{ eV})}$

$$E_{700 \text{ nm}} = 6.626 \times 10^{-34} \text{ J.s} \times \frac{3 \times 10^8 \text{ m/s}}{700 \times 10^{-9} \text{ m}}$$

$$= 2.84 \times 10^{-19} \text{ J} = 1.77 \text{ eV},$$

$$6. \quad \lambda_0 = \frac{h}{p}, \quad p = mV, \quad E = \frac{p^2}{2m}, \quad p = \sqrt{2mE} \Rightarrow \lambda_0 = \frac{h}{\sqrt{2m \times E}}$$

$$\lambda_0 = \frac{6.626 \times 10^{-34} \text{ J.s}}{\sqrt{2 \times 9.11 \times 10^{-31} \text{ kg} \times 12 \times 10^3 \text{ eV} \times 1.6 \times 10^{-19} \text{ J/eV}}} = 1.12 \times 10^{-11} \text{ m},$$

$\lambda_0 \approx 11.2 \text{ pm} \Rightarrow$ Electron microscope has a wavelength 4 orders of magnitude smaller than light microscope. A lower wavelength will result in a higher resolution of imaging. That means smaller samples can be observed with the electron microscope.

7. $kT \rightarrow J$
 temp. in K
 boltzman const

$$\begin{aligned} \text{---}40C^{\circ} \\ kT = 1.38 \times 10^{-23} \frac{m^2 kg}{s^2 K} \times 233 K = 3.2154 \times 10^{-21} J \\ = 2 \times 10^{-2} eV_{//} \end{aligned}$$

$$\begin{aligned} \text{---}85C^{\circ} \\ kT = 4.94 \times 10^{-21} J = 3.087 \times 10^{-2} eV_{//} \end{aligned}$$

$$\begin{aligned} \text{---}500C^{\circ} \\ kT = 1.067 \times 10^{-20} J = 6.67 \times 10^{-2} eV_{//} \end{aligned}$$

8. $I = \frac{\Delta Q}{\Delta t}$, $\Delta Q = -n.e$, $\Delta t = \frac{d}{V_{el}}$

a) $|I| = \frac{-ne V_{el}}{d}$

$$I = - \frac{n e V_{el}}{d}$$

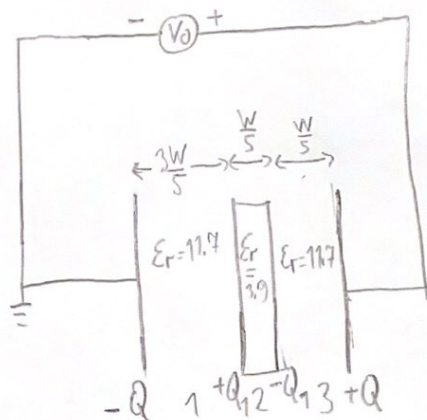
$\nwarrow$ # of e^- emitted
 $\nearrow$ e^- charge
 $\rightarrow$ ave. velocity of electrons
 $\rightarrow$ distance btwn plates

b) $I = - \frac{4 \times 10^5 \times 10^7 m/s \times 1.6 \times 10^{-19} C}{3 \times 10^{-6} m} = \boxed{-0.213 A_{//}}$

9) In thermoelectric generators which has a nonzero temperature gradient (one side of the conductor is hot and other side is cold) a current can be generated since heat flow will force charge carriers to move. But in the given setup conductor is in equilibrium meaning there is no heat flow. Electrons move in every direction with equal probability. Therefore, there will not be a net current conducted in the conductor

$$I_{net} = 0_{//}$$

10.



* Think of the setup as
3 capacitors connected in series

$$C_1 = 11.7 \frac{\epsilon_0 A}{\frac{3W}{5}} = C \quad C_2 = C \quad C_3 = 3C$$

$\downarrow$
 $19.5 \frac{\epsilon_0 A}{W}$

$$C_{eq} = \epsilon_0 \frac{A}{W} \left(\frac{3}{11.7 \times 5} + \frac{1}{3.9 \times 5} + \frac{1}{11.7 \times 5} \right)^{-1}$$

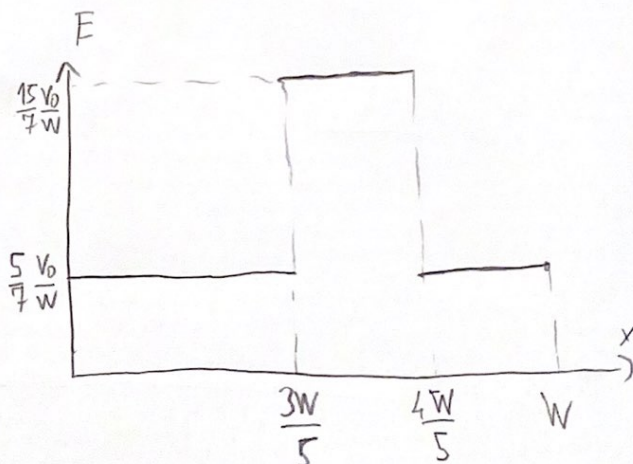
$$= 8.35 \frac{\epsilon_0 A}{W}$$

$$Q = V_0 C_{eq} = 8.35 \frac{\epsilon_0 A V_0}{W}$$

$$a) E_1 = \frac{Q/A}{\epsilon_r \epsilon_0} = \frac{8.35}{11.7} \frac{V_0}{W} = \frac{5}{7} \frac{V_0}{W}$$

$$E_2 = \frac{Q/A}{\epsilon_r \epsilon_0} = \frac{8.35}{3.9} \frac{V_0}{W} = \frac{15}{7} \frac{V_0}{W}$$

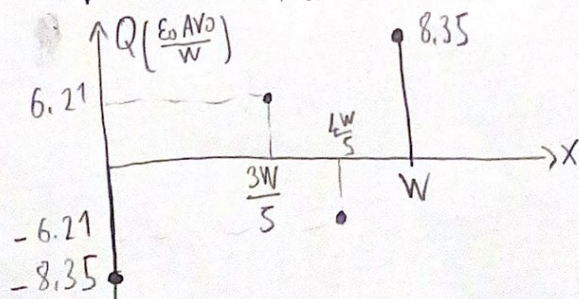
$$E_3 = \frac{Q/A}{\epsilon_r \epsilon_0} = \frac{8.35}{11.7} \frac{V_0}{W} = \frac{5}{7} \frac{V_0}{W}$$



$$b) P_2 = \epsilon_0 \chi_e E_2$$

$$\sigma_b = P_2 = (\epsilon_r - 1) \epsilon_0 E_2$$

$$Q_1 = 2.9 \epsilon_0 \frac{15}{7} \frac{V_0 A}{W} = 6.21 \frac{\epsilon_0 A V_0}{W}$$



$$c) V = - \int_{-\infty}^x E \cdot dx = \int_0^x E dx$$

